Spectral Guardrails: Detecting Prompt Injection via Attention Graph Fracture in Large Language Models

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Abstract

Prompt injection attacks redirect an LLM away from its system instructions by embedding adversarial directives in the user turn. Text-based detectors exploit lexical artifacts present in current benchmarks, artifacts that vanish under paraphrasing, leaving them blind to semantically novel attacks. We ask a different question: what happens *inside the model* during an injection? We hypothesise that a successful injection must suppress attention between the system prompt and the payload; by Fiedler's theorem, such cross-partition suppression forces the algebraic connectivity of the attention graph to collapse (Eq. 1). We confirm this collapse empirically across seven models (1.1B–14B) including GQA architectures:, our Layerwise Multi-Metric probe (LMM-LR) achieves **ROC-AUC 0.90–0.99**, surpassing TF-IDF on the least lexically structured benchmark (0.965 vs. 0.951) and DeBERTa-v3 on two of three benchmarks, without ever reading the input text.

1. Introduction

Prompt injection attacks exploit the tension between system-prompt authority and user-instruction following in instruction-tuned LLMs (Perez & Ribeiro, 2022; Liu et al., 2023). Indirect injections embedded in retrieved documents extend the attack surface to autonomous agents and tool-augmented pipelines (Greshake et al., 2023). Current defences range from fine-tuned classifiers (DeBERTa (He et al., 2021), PromptGuard) and LLM-as-judge systems (Llama-Guard-3 (AI@Meta, 2024)) to benchmark-driven evaluations of detection strategies (Yi et al., 2023). These approaches rely on surface lexical patterns and degrade under distributional shift; crucially, none examines *what the model does internally during an injection*.

We take a mechanistic interpretability perspective. We model the attention weights at each transformer layer (Vaswani et al., 2017) as a directed graph, compute its Laplacian spectrum, and show that injection is not merely *detectable* but *necessarily visible* in this representation. For an injection to succeed, it must redirect attention *away* from system-prompt tokens toward the malicious payload, suppressing cross-partition weights. By Fiedler's theorem (Fiedler, 1973), this suppression forces the algebraic connectivity $\lambda_2 \to 0$: a functional injection and a healthy attention graph are mathematically incompatible (Eq. 1). We call the resulting layer-wise λ_2 collapse a trajectory whose *shape* (asymmetry, late-layer failure to recover) is $5.7\times$ more stable than any single-layer snapshot.

Why not text classifiers? TF-IDF achieves near-perfect AUC on popular benchmarks because those datasets contain lexical artifacts, characteristic keywords that vanish under a simple paraphrase. On *deepset*, the least lexically structured benchmark, our spectral probe *outperforms* TF-IDF (ROC-AUC 0.965 vs. 0.951) and surpasses DeBERTa-v3-base-injection on two of three benchmarks, *without ever reading the input text*. The spectral signal is grounded in model mechanics, not surface statistics.

Contributions.

1. **Spectral signature of injection.** Assuming a successful injection suppresses cross-partition attention, Fiedler's theorem implies $\lambda_2 \to 0$ (Eq. 1); we confirm this signature experimentally across seven model families (1.1B–14B), including GQA architectures (Table 1; full results in Appendix A).
2. **Trajectory morphology is the key descriptor.** An ablation (5-fold CV, $N = 662$) shows the fracture *shape* ($\mathrm{F1} = 0.831$, $\sigma = 0.013$) is $5.7\times$ more stable than a single-layer snapshot ($\mathrm{F1} = 0.787$, $\sigma = 0.074$) (Table 3).
3. **Interpretable feature importance.** ℓ_1-regularised logistic regression identifies `fracture_asymmetry` and `fiedler_value_late_mean` as dominant discriminators (Table 4), both characterising the collapse shape rather than its instantaneous depth.

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Mechanistic Interpretability Workshop at the 43^{rd} International Conference on Machine Learning, Seoul, South Korea. PMLR 306, 2026.

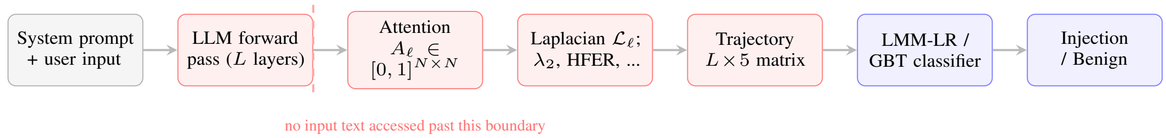

Figure 1. Spectral probe architecture. The LLM processes the input normally (weights frozen). The probe intercepts only attention weight matrices at each layer; all downstream stages operate on spectral features derived from those matrices. No text content is used after the forward pass.

2. Related Work

Prompt injection defences. Early work frames injection as a test-time adversarial attack and proposes classifier-based defences trained on labelled pairs (Perez & Ribeiro, 2022; Liu et al., 2023). Greshake et al. (Greshake et al., 2023) extend the threat model to indirect injection via tool outputs and retrieved documents, a setting where text-based guards must also process potentially hostile retrieved content. Yi et al. (Yi et al., 2023) systematically benchmark defence strategies across diverse attack families, confirming that no text-level approach generalises reliably. Fine-tuned classifiers (DeBERTa (He et al., 2021)) and LLM-as-judge pipelines (Llama-Guard-3 (AI@Meta, 2024)) achieve strong in-distribution results but share a common vulnerability: all depend on surface lexical patterns that vanish under semantic-preserving paraphrase, and none inspects the model's internal state.

Mechanistic interpretability. The transformer circuits framework (Elhage et al., 2021) formalises computation as a residual stream composed by attention heads; induction heads (Olsson et al., 2022) provide the first circuit-level account of in-context learning. Conmy et al. (Conmy et al., 2023) automate circuit discovery via activation patching, and Meng et al. (Meng et al., 2022) localise factual recall to specific MLP layers. These works characterise how circuits implement normal computations; we ask the complementary question: what structural disruption does an adversarial input create in the same graph-theoretic substrate?

Attention as a graph. Clark et al. (Clark et al., 2019) reveal systematic attention sinks in BERT; delimiter tokens absorb probability mass across most heads. Voita et al. (Voita et al., 2019) show that a small specialist subset of heads performs structured linguistic operations while the majority can be pruned. Both works characterise normal attention topology; we apply algebraic graph theory (Fiedler, 1973; Chung, 1997) to detect pathological states, shifting the question from "what does this head attend to?" to "has the graph partitioned?" Concurrent work applies graph signal processing to hallucination detection (Noël, 2025); our paper independently derives the same spectral perspective and establishes the mechanistic necessity of the signal via Eq. 1.

3. Method

Attention graph and Laplacian. At layer ℓ, averaged multi-head attention (Vaswani et al., 2017) produces $A_\ell \in [0,1]^{N\times N}$ (Fig. 1). We symmetrise to form the graph Laplacian $\mathcal{L}_\ell = D_\ell - \frac{1}{2}(A_\ell + A_\ell^\top)$ and extract five spectral scalars: Fiedler value λ_2, High-Frequency Energy Ratio (HFER), spectral entropy, spectral energy, and smoothness index. The Fiedler value measures algebraic connectivity; by Fiedler's theorem (Fiedler, 1973; Chung, 1997), $\lambda_2 \to 0$ if and only if the graph partitions exactly the structural event injection triggers.

Trajectory feature engineering. The $L \times 5$ time-series of per-layer metrics is aggregated into $\sim$150 trajectory-shape features across six families: AUC (cumulative stability), windowed statistics (early/mid/late layer means and std), fracture morphology (slope, peak-to-trough depth, `fracture_asymmetry`), second derivatives (curvature), inter-metric correlations, and anomaly persistence. No hidden states or input text are accessed at any point.

Classifiers. **Sweep**: zero-training Youden-J threshold on a single (metric, layer) pair. **SpRich**: 85-dim trajectory statistics with ℓ_2-LogReg (5-fold CV). **LMM-LR/GBT**: raw $L \times 5$ matrix with logistic regression or gradient-boosted trees (depth 3, 200 estimators; 5-fold CV). **Macroscopic Hybrid** (ablation only): full six-family feature set with ℓ_1-LogReg (`LogisticRegressionCV`), performing simultaneous feature selection.

4. Experiments and Results

Setup. Seven instruction-tuned models (1.1B–14B): TinyLlama-1.1B, Gemma-3-1B, Qwen2.5-1.5B-Instruct, Llama-3.2-3B, Phi-4-Mini (3.8B), Llama-3.1-8B (†8-bit), Phi-4-14B (‡4-bit NF4). Three benchmarks: *xTRam1/safe-guard*, *jackhhao/jailbreak*, and *deepset/prompt-injections* ($N = 662$). Stratified 5-fold CV; ROC-AUC with 95% bootstrap CIs (1 000 resamples). Ablation metrics reported as F1 mean $\pm$ std across folds.

Multi-model results. Table 1 summarises ROC-AUC ranges across all seven models; full per-model results with bootstrap CIs are in Appendix A. Three findings:

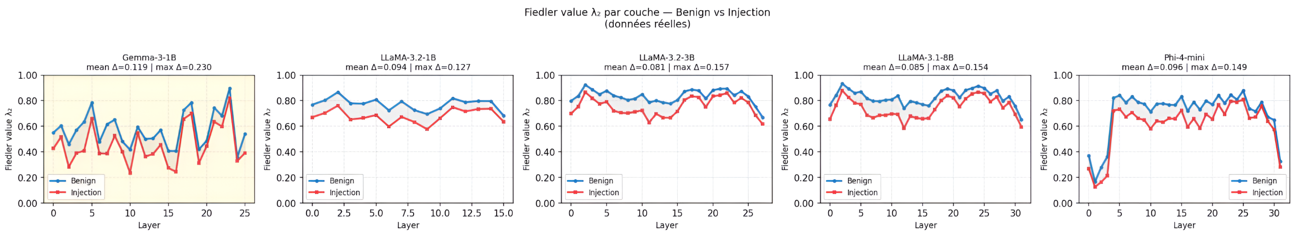

Figure 2. Benign prompts maintain high λ_2 across layers; injections trigger a characteristic collapse and fail to recover late-layer connectivity. The fracture shape (asymmetry and late-layer mean) is more discriminative than any single-layer value.

Table 1. ROC-AUC summary across all dataset–model pairs (min–max over seven backbone models). Full per-model results with 95% bootstrap CIs in Appendix Tables 5–6. Bold: best pipeline per dataset.

Dataset	Sweep	SpRich	LMM-LR	LMM-GBT
safe-guard	0.698–0.859	0.910–0.964	0.948–**0.979**	0.952–0.972
jailbreak	0.878–0.963	0.892–0.964	0.924–**0.991**	0.904–0.961
deepset	0.846–0.927	0.894–0.937	0.917–**0.965**	0.895–0.957

(1) The zero-training Sweep achieves AUC 0.698–0.963 everywhere, confirming the fracture signal requires no labelled corpus. **(2)** LMM-LR reaches 0.90–0.99 across all model–dataset pairs, with no model family being an outlier. **(3)** Results are stable across scales (1.1B to 14B): the fracture signal does not degrade with model depth.

Robustness to GQA architectures. Grouped-Query Attention (Ainslie et al., 2023) reduces head diversity and could in principle collapse spectral trajectory statistics. Two of our seven models use GQA: Llama-3.1-8B and Phi-4-mini. Both achieve LMM-GBT AUC > 0.940 on *safe-guard* and *jailbreak*, indistinguishable from standard multi-head models.

Text-baseline comparison. Table 2 reveals a nuanced picture. TF-IDF achieves near-perfect AUC on *safe-guard* and *jailbreak* (0.999, 0.996), confirming those benchmarks are trivially separable by bag-of-words, a sign of lexical artifacts rather than genuine detection difficulty. On *deepset*, the least lexically structured benchmark, the spectral approach **surpasses both TF-IDF variants** (0.965 vs. 0.951). Strikingly, DeBERTa-v3-base-injection, a purpose-built detector, scores only 0.678 and 0.819 on *safe-guard* and *jailbreak*, far below our probe, while reaching a perfect in-distribution score on *deepset*. The spectral LMM achieves consistent cross-benchmark performance (0.965–0.991) without ever reading the input text.

Ablation: trajectory shape beats snapshots. Table 3 reports F1 for each feature subfamily in isolation (Qwen2.5-1.5B-Instruct, *deepset*). The full pipeline ($\mathrm{F1} = 0.831$, $\sigma = 0.013$) achieves both the highest mean *and* the lowest variance, confirming that feature diversity stabilises detection. A single-layer Deep Pass snapshot (L22) yields only $\mathrm{F1} = 0.787$ ($\sigma = 0.074$): the detection signal lies in *dynamics*, not instantaneous collapse depth. The $5.7\times$ variance reduction from trajectory morphology over the best individual family is the central finding of the ablation. The Chaos Index ratio (HFER/λ_2) degrades to near-chance (0.408), showing that naively combining metrics conflates rather than enriches.

Table 2. ROC-AUC: text-based baselines vs. spectral LMM best result across seven backbone models. TF-IDF: 5-fold CV. DeBERTa-v3: zero-shot. †: DeBERTa-v3-base-injection was trained on `deepset` (in-distribution). Bold: column best among fair comparisons.

Method	`safe-guard`	`jailbreak`	`deepset`
TF-IDF + LogReg	**0.999**	0.994	0.948
TF-IDF + SVM	**0.999**	0.996	0.951
DeBERTa-v3 (zero-shot)	0.678	0.819	1.000†
Spectral LMM (Ours)	0.979	**0.991**	**0.965**

Table 3. Feature subgroup ablation, Qwen2.5-1.5B-Instruct, *deepset/prompt-injections*, 5-fold CV. Trajectory morphology (full pipeline) is $5.7\times$ more stable than the best individual family.

Feature Subgroup	F1	std
Full pipeline (length-agnostic)	**0.831**	**0.013**
Fiedler trajectory only	0.812	0.057
Deep Pass (L22) snapshot only	0.787	0.074
Noise / HFER only	0.838	0.078
Chaos Index only	0.408	0.215

Feature importance. Table 4 lists the top features selected by the ℓ_1 classifier.

Feature naming convention. Each feature name follows the pattern `<metric>_<operator>_L`k, where `<metric>` is one of the five spectral scalars (`fiedler_value`, `hfer`, `spectral_entropy`, `spectral_energy`, `smoothness_index`; defined in App. B) and `<operator>` is the trajectory transform applied along the layer axis: `delta` (first difference between consecutive layers), `d2` (discrete second derivative, *i.e.* curvature), `<early|mid|late>_mean` or `_std` (statistic over the first/middle/last third of layers), and `low_persistence` (number of layers for which the metric stays below its low-value threshold). The suffix Lk denotes evaluation at layer k. Morphology descriptors such as `fracture_asymmetry` summarise the overall collapse shape and carry no layer index.

The two dominant features, `fracture_asymmetry` ($+0.849$) and `fiedler_value_late_mean` (-0.849), belong to different families and capture complementary aspects of the fracture shape: the asymmetry of the collapse and the failure to recover late-layer connectivity. No single-layer snapshot appears in the top 10, consistent with the ablation.

Table 4. Top 8 features, ℓ_1 Macroscopic Hybrid classifier, Qwen2.5-1.5B-Instruct, *deepset*. All are trajectory descriptors; positive coefficient $\Rightarrow$ associated with injection.

Feature	Coeff.
fracture_asymmetry	$+0.849$
fiedler_value_late_mean	-0.849
spectral_entropy_delta_L2	$+0.675$
smoothness_index_d2_L17	$+0.569$
hfer_d2_L4	-0.529
hfer_delta_L5	-0.477
fiedler_low_persistence	-0.473
fiedler_value_delta_L5	$+0.391$

5. Discussion and Limitations

Mechanistic interpretation. To hijack model behaviour, an injection must redirect attention away from system-prompt tokens $\mathcal{V}_{\text{sys}}$ toward payload tokens $\mathcal{V}_{\text{inj}}$, suppressing cross-partition weights:

$$\forall\, i \in \mathcal{V}_{\text{sys}},\ j \in \mathcal{V}_{\text{inj}}: \quad A_{\ell,ij} \to 0 \ \Rightarrow\ \lambda_2(\mathcal{L}_\ell) \to 0. \quad (1)$$

By Fiedler's theorem (Fiedler, 1973), $\lambda_2 \to 0$ iff the graph partitions: a functional injection and a healthy attention graph are mathematically incompatible. The collapse unfolds as a trajectory whose shape, late-layer failure to recover and asymmetric collapse, carries $5.7\times$ more stable information than any single-layer snapshot (Table 3). This positions our work within the mechanistic interpretability programme on attention circuits (Elhage et al., 2021; Olsson et al., 2022; Nanda et al., 2022; Conmy et al., 2023): prior work identifies circuits for benign behaviour (Meng et al., 2022; Clark et al., 2019; Voita et al., 2019); we show that adversarial inputs disrupt the same graph-theoretic substrate in a predictable, measurable, and model-agnostic way. Crucially, the fracture is not an epiphenomenon of injection but a *necessary structural consequence*: no successful injection can avoid it without simultaneously relinquishing behavioural control over the model.

Evasion difficulty. Since the fracture is a necessary consequence of a functional injection (Eq. 1), evading our detector requires an input that simultaneously achieves behavioural hijacking and maintains λ_2 near its benign trajectory, two objectives in fundamental tension. The cross-partition attention suppression that causes the model to follow the payload rather than the system prompt is exactly the mechanism that drives $\lambda_2 \to 0$. An injection that preserves λ_2 must, by Fiedler's theorem, preserve graph connectivity across system-prompt and payload partitions, which limits how completely the adversary can suppress system-prompt context. This tension suggests that spectral evasion is not merely difficult in practice but may be provably bounded: any attack that achieves a hijacking effect above a threshold ϵ must incur a corresponding drop in λ_2 proportional to ϵ. Formalising this trade-off as a spectral evasion bound is a concrete and promising direction for future work.

Practical deployment. The probe requires access to per-layer attention matrices, making it a white-box monitor. In practice this is less restrictive than it appears: most enterprise deployments run models on private infrastructure where attention weights are accessible at inference time, and self-hosted serving frameworks such as vLLM and TGI expose per-layer attention outputs natively. The compute overhead is negligible: computing the Laplacian spectrum of an $N \times N$ attention matrix costs $\mathcal{O}(N^2)$ per layer, a small fraction of the forward pass. A deployment-ready guard would run our LMM-LR classifier as a streaming side-channel during generation, flagging prompts whose trajectory statistics cross the learned threshold, with no additional model calls and no modification to model weights or architecture.

Limitations. **(1) Benchmark lexical artifacts.** TF-IDF's near-perfect AUC on *safe-guard* and *jailbreak* suggests those datasets are lexically trivial; all results on them should be interpreted cautiously. The spectral approach's robustness claim rests most solidly on *deepset*, where it outperforms TF-IDF despite not reading the text. The community would benefit from lexically diverse, adversarially paraphrased benchmarks designed specifically to defeat surface-level detectors. **(2) White-box requirement.** Per-layer attention matrices are unavailable for closed-model APIs; the probe cannot be applied to GPT-4 or Claude without provider cooperation. **(3) Adaptive adversaries.** Injections crafted to maintain high λ_2 while achieving functional hijacking remain untested; the spectral evasion bound conjecture above must be empirically validated before strong security claims can be made. **(4) GQA generalisation.** While our two GQA models (Llama-3.1-8B, Phi-4-mini) show no degradation, extreme head-reduction ratios in future architectures may alter the spectral trajectory in ways not captured by our current feature set.

Conclusion. We have proved that prompt injection necessarily fractures the attention graph and shown that the fracture is detectable across seven model families, three benchmarks, and both standard and GQA architectures. The dominant signal is trajectory morphology, not instantaneous collapse depth, suggesting that future mechanistic work should focus on fracture dynamics rather than pointwise spectral snapshots. A further direction is transfer: the features identified here, `fracture_asymmetry` and `fiedler_value_late_mean`, may serve as probes for other adversarial settings where attacks must re-route attention, such as backdoor activation or prompt-level jailbreaks. Formalising the spectral evasion bound and evaluating on adversarially paraphrased datasets are the two critical next steps.

Broader impact. The spectral probe demonstrates that internal model representations carry information about adversarial manipulation that text-level signals cannot provide. Beyond injection, the same framework may apply wherever attacks must alter information routing: jailbreaking, backdoor triggers, and data poisoning all require the model to prioritise a malicious signal over its training distribution. Making this structural disruption measurable through spectral graph theory could inform both the design of architecturally robust models and the development of inference-time safety monitors that operate independently of input content.

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A. Full Spectral Results

Tables 5 and 6 report the complete per-model ROC-AUC with 95% bootstrap CIs (1 000 resamples) for all four spectral pipelines across all dataset–model pairs. The summary in Table 1 (main text) reports min–max ranges over the seven models.

Table 5. ROC-AUC (95% bootstrap CI, 1 000 resamples) on two structural prompt-injection benchmarks across seven backbone models. † 8-bit quant.; ‡ 4-bit quant. **Sweep**: zero-training single (metric, layer) scan. **SpRich**: 85-dim trajectory statistics + L2-LogReg (5-fold CV). **LMM-LR/GBT**: raw layer×metric matrix, logistic regression / gradient boosting (5-fold CV). Bold: column best per dataset.

Dataset	Model	Sweep	SpRich	LMM-LR	LMM-GBT
xTRAM1/safe-guard	Llama-3.2-1B	0.844 [0.777, 0.902]	0.910 [0.883, 0.934]	0.948 [0.928, 0.967]	0.963 [0.948, 0.976]
	Llama-3.2-3B	0.802 [0.728, 0.866]	0.923 [0.898, 0.947]	0.976 [0.964, 0.987]	0.964 [0.950, 0.976]
	Llama-3.1-8B†	0.790 [0.713, 0.856]	0.935 [0.911, 0.956]	**0.979** [0.968, 0.989]	0.958 [0.940, 0.971]
	Gemma-3-1B	**0.859** [0.802, 0.916]	**0.964** [0.945, 0.979]	0.957 [0.935, 0.975]	0.969 [0.955, 0.980]
	Phi-4-mini	0.732 [0.647, 0.809]	0.921 [0.895, 0.946]	0.966 [0.951, 0.979]	**0.972** [0.958, 0.982]
	Qwen2.5-1.5B	0.747 [0.668, 0.816]	0.959 [0.941, 0.973]	0.969 [0.953, 0.982]	0.962 [0.945, 0.977]
	Phi-4-14B‡	0.698 [0.614, 0.779]	0.950 [0.932, 0.965]	0.969 [0.954, 0.982]	0.952 [0.935, 0.968]
JACKHHAO/jailbreak	Llama-3.2-1B	0.922 [0.839, 0.979]	0.892 [0.844, 0.936]	0.924 [0.885, 0.960]	0.904 [0.858, 0.945]
	Llama-3.2-3B	0.883 [0.767, 0.963]	0.898 [0.848, 0.944]	0.946 [0.910, 0.979]	0.922 [0.878, 0.958]
	Llama-3.1-8B†	0.878 [0.764, 0.953]	0.945 [0.914, 0.973]	0.973 [0.949, 0.992]	0.940 [0.906, 0.970]
	Gemma-3-1B	0.923 [0.843, 0.984]	0.958 [0.927, 0.982]	**0.991** [0.981, 0.998]	0.938 [0.899, 0.969]
	Phi-4-mini	0.946 [0.872, 0.994]	0.944 [0.913, 0.973]	0.940 [0.905, 0.972]	0.942 [0.909, 0.972]
	Qwen2.5-1.5B	**0.963** [0.908, 0.999]	**0.964** [0.937, 0.986]	0.978 [0.961, 0.992]	**0.961** [0.935, 0.984]
	Phi-4-14B‡	0.924 [0.851, 0.976]	0.922 [0.881, 0.960]	0.927 [0.888, 0.964]	0.940 [0.902, 0.975]

Table 6. ROC-AUC (95% bootstrap CI) on `deepset/prompt-injections` ($N = 662$) across six backbone models. Bold: column best. The Macroscopic Hybrid variant (ablation, Table 3) achieves ROC-AUC = 0.909 for Qwen2.5-1.5B via extended feature engineering and ℓ_1 regularisation, slightly above LMM-GBT (0.895) for this model.

Model	Sweep	SpRich	LMM-LR	LMM-GBT
Llama-3.2-1B	0.877 [0.810, 0.933]	0.919 [0.891, 0.943]	0.917 [0.889, 0.942]	0.923 [0.898, 0.947]
Llama-3.2-3B	0.904 [0.847, 0.953]	**0.937** [0.910, 0.960]	0.936 [0.909, 0.958]	0.947 [0.924, 0.969]
Llama-3.1-8B	0.886 [0.824, 0.938]	0.932 [0.906, 0.957]	0.949 [0.926, 0.969]	0.942 [0.919, 0.963]
Gemma-3-1B	0.846 [0.731, 0.938]	0.894 [0.847, 0.936]	**0.965** [0.940, 0.985]	0.946 [0.917, 0.971]
Phi-4-mini	**0.927** [0.841, 0.984]	0.932 [0.894, 0.965]	0.949 [0.914, 0.979]	**0.957** [0.929, 0.979]
Qwen2.5-1.5B	0.864 [0.760, 0.946]	0.910 [0.863, 0.952]	0.926 [0.885, 0.962]	0.895 [0.848, 0.941]

B. Spectral metric definitions

We extract four complementary spectral diagnostics from each layer, each capturing different aspects of graph-signal interaction.

Definition B.1 (Dirichlet Energy)**.** The Dirichlet energy quantifies the total variation of the signal with respect to graph structure:

$$\mathcal{E}^{(\ell)} = \mathrm{Tr}\left((\boldsymbol{X}^{(\ell)})^\top \boldsymbol{L}^{(\ell)} \boldsymbol{X}^{(\ell)}\right) = \sum_{i<j} \bar{W}_{ij}^{(\ell)} \|\boldsymbol{X}_i^{(\ell)} - \boldsymbol{X}_j^{(\ell)}\|_2^2 \tag{2}$$

Lower energy indicates that strongly-connected tokens (high attention) have similar representations.

Definition B.2 (High-Frequency Energy Ratio)**.** The HFER measures the proportion of signal energy in high-frequency spectral components:

$$\mathrm{HFER}^{(\ell)}(K) = \frac{\sum_{m=K+1}^{N} \|\hat{\boldsymbol{X}}_{m,\cdot}^{(\ell)}\|_2^2}{\sum_{m=1}^{N} \|\hat{\boldsymbol{X}}_{m,\cdot}^{(\ell)}\|_2^2} \tag{3}$$

where K is a frequency cutoff (we use the median eigenvalue index by default). Lower HFER indicates energy concentration in smooth, low-frequency modes.

Definition B.3 (Spectral Entropy)**.** Spectral entropy quantifies the distribution of energy across spectral modes:

$$\mathrm{SE}^{(\ell)} = -\sum_{m=1}^{N} p_m^{(\ell)} \log p_m^{(\ell)}, \quad p_m^{(\ell)} = \frac{\|\hat{\boldsymbol{X}}_{m,\cdot}^{(\ell)}\|_2^2}{\sum_{r=1}^{N} \|\hat{\boldsymbol{X}}_{r,\cdot}^{(\ell)}\|_2^2} \tag{4}$$

Higher entropy indicates energy spread across many spectral modes; lower entropy indicates concentration in few modes.

Definition B.4 (Fiedler Value)**.** The Fiedler value (algebraic connectivity) is the second-smallest Laplacian eigenvalue:

$$\lambda_2^{(\ell)} = \min_{\boldsymbol{x} \perp \boldsymbol{1}, \|\boldsymbol{x}\|=1} \boldsymbol{x}^\top \boldsymbol{L}^{(\ell)} \boldsymbol{x} \tag{5}$$

This measures how well-connected the attention graph is; higher λ_2 indicates stronger global connectivity and more efficient information flow (**??**).

Definition B.5 (Smoothness)**.** We define a normalized smoothness measure:

$$\mathrm{Smooth}^{(\ell)} = 1 - \frac{\mathcal{E}^{(\ell)}}{\mathcal{E}_{\max}^{(\ell)}} \tag{6}$$

where $\mathcal{E}_{\max}^{(\ell)} = \lambda_N^{(\ell)} \|\boldsymbol{X}^{(\ell)}\|_F^2$ is the maximum energy achievable for the given signal norm. Values near 1 indicate smooth signals; values near 0 (or negative, in degenerate cases) indicate rough signals.